Similarly, $\tan^{-1}\left(\frac{d}{1+a_2a_3}\right) = \tan^{-1}\left(\frac{a_3-a_2}{1+a_2a_3}\right) = \tan^{-1}a_3 - \tan^{-1}a_2$.

Continuing inductively, we get

$$\tan^{-1}\left(\frac{d}{1+a_na_{n-1}}\right) = \tan^{-1}\left(\frac{a_n-a_{n-1}}{1+a_{n-1}a_n}\right) = \tan^{-1}a_n - \tan^{-1}a_{n-1}.$$

Adding vertically, we get

$$\tan^{-1}\left(\frac{d}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{d}{1+a_2a_3}\right)+\ldots+\tan^{-1}\left(\frac{d}{1+a_na_{n-1}}\right) = \tan^{-1}a_n - \tan^{-1}a_1.$$

$$\tan\left[\tan^{-1}\left(\frac{d}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{d}{1+a_2a_3}\right)+\ldots+\tan^{-1}\left(\frac{d}{1+a_na_{n-1}}\right)\right] = \tan[\tan^{-1}a_n - \tan^{-1}a_1]$$

$$= \tan\left[\tan^{-1}\left(\frac{a_n-a_1}{1+a_la_n}\right)\right] = \frac{a_n-a_1}{1+a_la_n}.$$

Example 4.24 Solve $\tan^{-1}\left(\frac{1-x}{1+x}\right)=\frac{1}{2}\tan^{-1}x$ for $x>0$.

Solution

$\tan^{-1}\left(\frac{1-x}{1+x}\right)=\frac{1}{2}\tan^{-1}x$ gives $\tan^{-1}1-\tan^{-1}x=\frac{1}{2}\tan^{-1}x$.

Therefore, $\frac{\pi}{4}=\frac{3}{2}\tan^{-1}x$, which in turn reduces to $\tan^{-1}x=\frac{\pi}{6}$

Thus, $x=\tan\frac{\pi}{6}=\frac{1}{\sqrt{3}}$.

Example 4.25 Solve $\sin^{-1}x>\cos^{-1}x$

Solution

Given that $\sin^{-1}x>\cos^{-1}x$. Note that $-1\le x\le 1$.

Adding both sides by $\sin^{-1}x$, we get

$\sin^{-1}x+\sin^{-1}x>\cos^{-1}x+\sin^{-1}x$, which reduces to $2\sin^{-1}x>\frac{\pi}{2}$.

As sine function increases in the interval $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$, we have $x>\sin\frac{\pi}{4}$ or $x>\frac{1}{\sqrt{2}}$.

Thus, the solution set is the interval $\left(\frac{1}{\sqrt{2}},1\right]$.

Example 4.26

Show that $\cot(\sin^{-1}x)=\frac{\sqrt{1-x^2}}{x}, -1\le x\le 1$ and $x\neq 0$

Solution

Let $\sin^{-1}x=\theta$. Then, $x=\sin\theta$ and $x\neq 0$, we get $\theta\in\left[\frac{-\pi}{2},0\right)\cup\left(0,\frac{\pi}{2}\right]$.

Hence, $\cos\theta\ge 0$ and $\cos\theta=\sqrt{1-\sin^2\theta}=\sqrt{1-x^2}$.

Thus, $\cot(\sin^{-1}x)=\cot\theta=\frac{\sqrt{1-x^2}}{x}$, $|x|\le 1$ and $x\neq 0$

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